ORLANDO HEALTH®

1414 Kuhl Ave.
Orlando, FL 32806

DEPARTMENT GUIDELINE

Guideline Title: **Pediatric Supraventricular Tachycardia**	Guideline Number: **GUID-3203-PED017**

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature:	Signature:
Department Numbers: 3203	

I. **PURPOSE:** Ensure prompt identification and provide intervention to restore a stable cardiac rhythm with adequate cardiac output and.

II. **PATHOPHYSIOLOGY:**

Supraventricular tachycardia (SVT) is a regular, rapid rhythm that arises from either electrical re-entry or an ectopic focus. Reentrant tachycardia is the most common. SVT can occur in a normal heart, or in association with rheumatic heart disease, pericarditis, physiologic disturbance, mitral valve prolapse, or a pre-excitation syndrome. Ectopic SVT may be seen in patients with chronic lung disease, pneumonia, and poisoning. SVT is the most common significant dysrhythmia in pediatric patients. It is often well tolerated. The loss of atrial contribution to cardiac output, coupled with decreased ventricular filling time, contributes to lower cardiac output and the resultant symptoms of the SVT.

- Infant heart rates are usually >220 beats/minute.
- Children heart rates are usually >180 beats/minute.
- P waves absent/abnormal and RR interval not variable

III. **DEFINITIONS:**

A. SVT – Supraventricular tachycardia

IV. **DEPARTMENT GUIDELINE:**

A. Assess and manage airway, breathing, and circulation as per *GUID3203-PED001 – Pediatric General Management*.

B. Establish IV/IO access.

C. If signs of poor perfusion, administer 0.9% NS or Lactated Ringers 20 ml/kg IV, may be repeated per *GUID3203-PR015 – Intravenous Fluid Therapy*.

D. Obtain a 12-lead ECG if available.

E. Identify and treat possible causes of tachycardia: hypoxemia, hypovolemia, hyperthermia, metabolic disorders, cardiac tamponade, toxins, and pain.

F. <u>**STABLE SVT:**</u>

a. Attempt vagal maneuvers.

i. Place a bag with ice water over the upper half of an infant’s face (without obstructing the airway) for 15 to 20 seconds.

ii. Ask older children to bear down, blow through an obstructed straw, or try to push the plunger out of a 10 mL syringe by blowing into the tip.

iii. If the child is stable and the rhythm does not convert, you may repeat the attempt. If the second attempt fails, select another method or provide medication therapy. If the child is unstable, attempt vagal maneuvers only while preparing for pharmacologic or electrical cardioversion.

b. Consider treating pain and/or anxiety per *GUID3203-G005 - Analgesia and Sedation Management*.

c. **Adenosine 0.1 mg/kg rapid IV/IO push** with flush (max dose 6 mg)

   i. If the dysrhythmias persist after 1-2 minutes**- Adenosine may be repeated at 0.2 mg/kg IV** (max dose 12mg)

G. Unstable SVT:

a. A trial of adenosine is appropriate only if there is no delay to therapy

b. Consider pre-medication treat pain and/or anxiety per *GUID3203-G005 - Analgesia and Sedation Management.*

   a. Synchronized cardiovert per *GUID3203-3203 - Cardioversion.*

      i. Synchronized cardiovert at 0.5 J/kg. Reassess rhythm.

      ii. Synchronized cardiovert at 1 J/kg. Reassess Rhythm.

      iii. Synchronized cardiovert at 2 J/kg. Reassess Rhythm, repeat as necessary.

   b. Amiodarone may be considered when treating hemodynamically unstable SVT refractory to vagal maneuvers, adenosine, and cardioversion and for whom expert consultation is not available. Use with caution if hepatic failure is present. **Amiodarone 5mg/kg IV over 20-60 minutes** (max single dose: 150 mg). Monitor for hypotension.

   c. If no response to therapy, contact Medical Control or seek expert opinion regarding Procainamide or Verapamil for children < 1 year old.

H. Always consider the latest recommendations from the American Heart Association.

V. **DOCUMENTATION:**

A. EMS Charts

VI. **REFERENCES:**

A. Pediatric Advanced Life Support Manual, 2020. American Heart Association.